How-To Library

Informatica

Configuring the Simba ODBC Driver for MongoDB

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Abstract

You can configure an ODBC connection to connect to MongoDB from the Developer Tool or PowerCenter. This article explains how to configure an ODBC connection for MongoDB in the Developer Tool or PowerCenter.

Supported Versions

- Informatica 10.2

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Simba MongoDB ODBC Driver Configuration Overview

Configure the Simba MongoDB ODBC driver to connect to MongoDB from the Developer Tool or PowerCenter®. You can configure the Simba MongoDB ODBC driver on Windows or Linux.

To configure the Simba MongoDB ODBC driver, perform the following tasks:

1. Configure the Simba MongoDB ODBC driver.
2. Configure the MongoDB ODBC data source name.
3. Connect to MongoDB from the Schema Editor.

Prerequisites

Before you configure the Simba MongoDB ODBC driver, you must complete the following prerequisites:

- Install or upgrade to Informatica® services 10.2.
- Ensure that you have the PowerExchange® for MongoDB license file. You do not require a separate ODBC license to use PowerExchange for MongoDB.
- On Windows, download and install the Microsoft Visual C++ 2013 Redistributable Package in server and client machines from the Microsoft website. For example, download the vc_redist_x86.exe file.
- Ensure that you have applied the Informatica EBF-9648 on Informatica services 10.2.

  Note: For more information, contact Informatica Global Customer Support.

Configuring the Simba MongoDB ODBC Driver on Linux

You must configure the Simba MongoDB ODBC driver before you can run MongoDB mappings in Developer Tool or MongoDB sessions and workflows in PowerCenter.

Configuring the Data Source Name on Linux

1. Add the DSN entry for the MongoDB data source in the `odbc.ini` file.

   The following section shows a sample entry in the `odbc.ini` file:

   ```
   [Sample_Mongodb_ODBC_DSN]
   Description=Informatica MongoDB ODBC Driver (64-bit) DSN
   Driver=/export/home/adputf_9/Informatica/10.2/tools/mongodb/lib/
   libinformaticamongodbodbc64.so
   Server=HOST_NAME
   Port=27017
   AuthMechanism=SCRAM-SHA-1
   Database=TEST
   UID=USER_NAME
   PWD=PASSWORD
   AuthSource=admin
   ```

2. Set the ODBCINI environment variable on the Linux machine to the location of the `odbc.ini` file in which you created the data source name.

   To set the ODBCINI environment variable, you must export the ODBCINI variable.

   For example, run the following script from the Linux machine: export `ODBCINI=<Informatica installation directory>/odbc.ini`

3. Set LD_LIBRARY_PATH on the Linux machine to the following location: `<Informatica installation location>/export/home/adputf_9/Informatica/10.2/tools/mongodb/lib/`

   To set LD_LIBRARY_PATH, you must export the LD_LIBRARY_PATH variable.

   For example, run the following script from the Linux machine: export `LD_LIBRARY_PATH="${LD_LIBRARY_PATH}:<Informatica installation directory>/export/home/adputf_9/Informatica/10.2/tools/mongodb/lib/`

4. Restart the Integration Service.

Configuring the Schema Editor on Linux

To ensure that the Schema Editor works on the Linux machine where the Informatica Services is installed, you must set the ODBCSYSINI environment variable.

Configure the following steps to set the ODBCSYSINI environment variable:

1. Add the following MongoDB ODBC driver entry in the `odbcinst.ini` file:

   ```
   [ODBC Drivers]
   Informatica MongoDB ODBC Driver 64-bit=Installed
   [Informatica MongoDB ODBC Driver 64-bit]
   Description=Informatica MongoDB ODBC Driver (64-bit)
   Driver=/home/adputf_9/Informatica/10.2/tools/mongodb/lib/libinformaticamongodbodbc64.so
   ```

2. Set the ODBCSYSINI environment variable on the Linux machine to the location of the `odbcinst.ini` file.
   To set the ODBCSYSINI environment variable, you must export the ODBCSYSINI variable.

   For example, run the following script from the Linux machine:

   ```
   export ODBCSYSINI=<Informatica installation directory>/odbcinst.ini
   ```

3. Restart the Integration Service.

Connection Configuration Preferences for Schema Editor

The driver searches for the environment variables ODBCINI and ODBCSYSINI to locate the configuration files and read the DSN.

If the environment variables do not exist, the schema editor searches for the files in the following directories:

- `/etc/odbc.ini`
- `/etc/odbcinst.ini`

Configuring the Simba MongoDB ODBC Driver on Windows

Create the required 64-bit DSN for MongoDB using the Informatica MongoDB driver registry.

Configuring the Data Source Name on Windows

Configure the data source name before you create an ODBC connection to connect to MongoDB. You must configure the MongoDB ODBC driver properties, advanced properties, and schema when you configure a data source name.

Create a data source name in the ODBC datasource administrator to extract data from and load data to a MongoDB database. The data source properties provide information for the MongoDB server and the database. The advanced properties are read and write operations. You can also define a schema after you create a database.

You can find the ODBC datasource administrator in the Control Panel on Windows. Access the 64-bit ODBC datasource administrator in the machine where you installed Informatica Services and clients.

MongoDB ODBC Data Source Properties

You must configure a MongoDB ODBC data source before you can import MongoDB data sources. You can configure the data source properties on the **Simba MongoDB ODBC Driver DSN Setup** page.

The following table describes the MongoDB ODBC data source properties:

Property	Description
Data Source Name	Name of the data source name.
Description	Description to identify a data source name.
Server	Server name of the MongoDB server.
Port	Port from which you can access the MongoDB server.
Database	MongoDB database in the server that you want to access.
Connect to Replica Set	The replica set in the MongoDB implementation. Select if you want to connect to the replica set in MongoDB.
Replica Set Name	Name of the replica set of the database.
Secondary Servers	A comma-separated list of the servers in the replica set. You can specify the TCP port that each server uses to listen for client connections by appending a colon (:) and the port number to the server name or IP address.
Read Preference	Specifies how the driver routes read operations to the members of a replica set.

Property	Description
Authentication Mechanism	The authentication mechanism that the driver uses to connect to the MongoDB server.
Service Name	The service name of the MongoDB server.
Authentication Source	Name of the database to which you want to connect.
Username	The user name for accessing the MongoDB database.
Password	The password corresponding to the user name for the MongoDB database.
Encrypt Password	Encrypts credentials. You can choose one of the following options: - Current User Only. Select if the credentials are used only by the current Windows user. - All Users of this Machine. Select if the credentials are used by all users on the current Windows machine.

MongoDB ODBC Data Source Advanced Properties

When you create a data source name, configure the advanced properties from **Advanced Options**.

The following table describes the advanced properties in the MongoDB ODBC driver:

Property	Description
Enable Double-Buffering	Specifies whether the driver retrieves the data using double-buffering or single-buffering. When enabled, the driver uses double-buffering to retrieve the data. When disabled, the driver uses single-buffering to retrieve the data.
Documents to fetch per block	The maximum number of documents that a query returns at a time. Default is 4096.
Expose strings as SQL_WVARCHAR	Specifies how the String data type maps to SQL. When enabled, the MongoDB String data type maps to SQL_WVARCHAR. When disabled, the MongoDB String data type maps to SQL_VARCHAR.
String Column Size	The maximum number of characters that a STRING column can contain. You can specify a maximum value of 2147483647.
Expose binary as SQL_LONGVARBINARY	Specifies whether the driver returns Binary columns as data of type SQL_LONGVARBINARY or SQL_VARBINARY. When enabled, the driver returns Binary columns as SQL_LONGVARBINARY data. When disabled, the driver returns Binary columns as SQL_VARBINARY data.
Binary Column Size	The maximum data length for Binary columns. You can specify a maximum value of 2147483647.

Property	Description
Enable Passdown	Specifies whether the driver optimizes joins between virtual tables, and passes filtering and aggregation optimizations to the MongoDB database for handling. When enabled, the driver optimizes joins between virtual tables, and passes filtering and aggregation optimizations to the MongoDB database for handling. When disabled, the driver does not optimize joins, and does not pass filtering and aggregation optimizations to the MongoDB database.
Mechanism	Specifies where the driver looks for the schema definition. If you select Database, the driver loads the schema definition from the MongoDB database. If you select Local File, the driver loads the schema definition from the JSON file specified in the **Local File** field or the LocalMetadataFile key.
Sampling Method	Specifies how the driver samples data when generating a temporary schema definition. You can choose from the following options: - Forward: The driver samples data starting from the first record in the database, then samples the next record, and so on. - Backwards: The driver samples data starting from the last record in the database, then samples the preceding record, and so on.
Sampling Count	The maximum number of records that the driver can sample to generate the schema definition. To sample every record in the database, set this option to 0.
Sampling Interval	The interval at which the driver samples records when scanning through the database to generate a temporary schema definition. For example, if you set the sampling interval to 2, the driver samples every second record in the database.

Connect to MongoDB from the Schema Editor

Use the Schema Editor to connect to MongoDB through the Simba MongoDB ODBC driver.

Use one of the following options to connect to MongoDB from the Schema Editor when you want to define or edit the metadata:

DSN List

If you have created the DSN on Windows or Linux, the Schema Editor lists the DSN in the DSN list using the Informatica MongoDB 64-bit ODBC driver.

The Schema Editor uses the following approach to use the DSN on Windows or Linux:

- On Windows. When the Schema Editor prompts you to provide the connection information, select the DSN you created for MongoDB from the DSN list. The editor uses the specified DSN to connect to MongoDB.
- On Linux. The Schema Editor searches for the `odbc.ini` file specified by the ODBCINI environment variable to read the DSN. If the ODBCINI environment variable is not available, the Schema Editor reads the DSN from the `/etc/odbc.ini` file.

Connection String

The Schema Editor recognises the following format of the connection string to connect to MongoDB:

```
Driver={Informatica MongoDB ODBC Driver 64-
bit};Server={HOST_NAME};Port={27017};AuthMechanism={SCRAM-
SHA-1};UID={USER_NAME};PWD={PASSWORD};Database={DATABASE};AuthSource={AUTH_DATABASE}
```

The Schema Editor uses the following approach to use the connection string:

- On Windows, the driver name in the connection string is the name of the MongoDB registry.
- On Linux, the name of the driver is the name specified in the `odbcinst.ini` file for MongoDB.

Note: When you configure the Schema Editor to connect to MongoDB, the Schema Editor sometimes displays the following error message: `"The driver specified is not supported."`

Consider the following solution to connect to MongoDB:

1. Rename the MongoDB driver library file name.
   - On a Windows 64-bit system, rename the MongoDB driver library file name from `SimbaMongoDBODBC64.dll` to `MongoDBODBC_sb64.dll`
   - On a Windows 32-bit system, rename the MongoDB driver library file name from `SimbaMongoDBODBC32.dll` to `MongoDBODBC_sb32.dll`
   - On a Linux 64-bit system, rename the MongoDB driver library file name from `libinformaticamongodbodbc64.so` to `libmongodbodbc_sb64.so`
2. Using the administrator user account for your operating system, update the registry to reflect the changes made to the MongoDB driver library file name.
   - On a Windows 64-bit system, update `"Driver"="<Informatica installation directory>\clients\tools\mongodb\lib\MongoDBODBC_sb64.dll"` and `"Setup"="<Informatica installation directory>\clients\tools\mongodb\lib\MongoDBODBC_sb64.dll"` in the following MongoDB registry:

     ```
     [HKEY_LOCAL_MACHINE\SOFTWARE\ODBC\ODBCINST.INI\Informatica MongoDB ODBC Driver]
     ```
   - On a Windows 32-bit system, update `"Driver"="<Informatica installation directory>\clients\tools\mongodb\32bit\lib\MongoDBODBC_sb32.dll"` and `"Setup"="<Informatica installation directory>\clients\tools\mongodb\32bit\lib\MongoDBODBC_sb32.dll"` in the following MongoDB registry:

     ```
     [HKEY_LOCAL_MACHINE\SOFTWARE\Wow6432Node\ODBC\ODBCINST.INI\Informatica MongoDB ODBC Driver]
     ```
   - On a Linux 64-bit system, update the driver attribute in the `odbcinst.ini` file:

     ```
     Driver=<Informatica installation directory>\clients\tools\mongodb\lib\libmongodbodbc_sb64.so
     ```

     You must also update the driver attribute in the DSN entry for the MongoDB data source in the `odbc.ini` file.

Rules and Guidelines to Connect to MongoDB

Consider the following rules and guidelines when you use the Simba MongoDB ODBC driver:

- You cannot use the Simba MongoDB ODBC driver to read or write MongoDB column or table names that have more than 128 characters.
- To use the JSON filter in the schema editor, you must specify the filter in the following format: `{"_id":"57e1a9f855a1664555289b54"}`

Third-Party Known Limitations

The following table lists the third-party known limitations:

Bug	Description
OCON-9730	The Simba MongoDB ODBC driver fails to support the _ID colum, if present in an array element or nested document. Simba MongoDB ODBC driver reference number: 00093670
OCON-9718	The Simba MongoDB ODBC driver treats the MongoDB Views as read only tables. Simba MongoDB ODBC driver reference number: 00094915
OCON-9710	The Simba MongoDB ODBC driver does not support query override feature when parent table and virtual table are used together in a MongoDB source.
OCON-9706	The Simba MongoDB ODBC driver does not support Unicode metadata. Simba MongoDB ODBC driver reference number: 00094913
OCON-9699	When you select the **Refresh Schema** option for a virtual table in a MongoDB source and use the **Mapping flow** option to write data to a virtual table in a MongoDB target, the mapping fails.
OCON-9697	The Simba MongoDB ODBC driver fails to create the target table as the driver does not support **Drop Table** and **Create Table** commands.
OCON-9695	When you refresh a schema for a MongoDB connection in the Connection Explorer view of the Developer tool or in PowerCenter, the Simba MongoDB ODBC driver fails to refresh the schema in the table. Workaround: Disconnect the connection and re-connect to list the table. Simba MongoDB ODBC driver reference number: 00094912
CON-8147	When you use the Simba MongoDB ODBC driver, you cannot read or write records in JSON format. Simba MongoDB ODBC driver reference number: 00094516
CON-7530	When you use the Simba MongoDB ODBC driver to connect to MongoDB, you cannot truncate the target table. Simba MongoDB ODBC driver reference number: 00092615
CON-2347	You cannot use Mongo shell commands in preprocessing and postprocessing commands, and in saved and custom queries of a task to write data to MongoDB. Simba MongoDB ODBC driver reference number: 00094260

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